Barcelona School of Economics

Master in Economics

Shock propagation through the lens of remittances

Who bears the cost? Evidence from US-Mexico migration corridors

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ABSTRACT IN ENGLISH

This paper studies how localized shocks in migrant destination areas propagate to origin communities through remittance networks. We use Hurricane Ian, which struck Florida in 2022, as a spatially concentrated negative shock to remittance senders in the United States. Combining municipality-level remittance data from Banco de México with migration-network measures based on Mexican consular records, we examine whether Mexican municipalities more exposed to Florida experienced differential changes in remittance inflows after the shock. We then use the structure of migrant networks to study whether exposure to other major U.S. destinations, especially Texas and California, mitigated or amplified the shock. The results speak to how international remittance networks transmit local economic disruptions across borders and to the role of migrant-network diversification in shaping the distribution of these effects.

ABSTRACT IN CATALAN

Aquest treball investiga com els xocs localitzats a les àrees de destinació dels migrants es propaguen cap a les comunitats d'origen a través de les xarxes de remeses. Utilitzem l'huracà Ian, que va colpejar Florida l'any 2022, com un xoc negatiu espacialment concentrat sobre els emissors de remeses als Estats Units. Combinant dades de remeses a escala municipal del Banc de Mèxic amb mesures de xarxes migratòries basades en registres consulars mexicans, examinem si els municipis mexicans amb una exposició més gran a Florida van experimentar canvis diferencials en les entrades de remeses després del xoc. A continuació, utilitzem l'estructura de les xarxes migratòries per estudiar si l'exposició a altres grans destinacions dels Estats Units, especialment Texas i Califòrnia, va mitigar o amplificar el xoc. Els resultats aporten evidència sobre com les xarxes internacionals de remeses transmeten pertorbacions econòmiques locals a través de les fronteres i sobre el paper que té la diversificació de les xarxes migratòries en la distribució d'aquests efectes.

KEYWORDS IN ENGLISH : remittances, migration networks, localized shocks

KEYWORDS IN CATALAN : remeses, xarxes migratòries, xocs localitzats

MASTER PROJECT

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Abstract

This paper studies how localized shocks in migrant destination areas propagate to origin communities through remittance networks. We use Hurricane Ian, which struck Florida in 2022, as a spatially concentrated negative shock to remittance senders in the United States. Combining municipality-level remittance data from Banco de México with migration-network measures based on Mexican consular records, we examine whether Mexican municipalities more exposed to Florida experienced differential changes in remittance inflows after the shock. We then use the structure of migrant networks to study whether exposure to other major U.S. destinations, especially Texas and California, mitigated or amplified the shock. The results speak to how international remittance networks transmit local economic disruptions across borders and to the role of migrant-network diversification in shaping the distribution of these effects.

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Contents

1. Introduction

International remittances are one of the main private insurance channels through which local economic shocks can cross national borders (Amuedo-Dorantes & Pozo, 2006; Bettin et al., 2025; Yang & Choi, 2007). In 2024, Mexico received more than US$67 billion in personal remittances, making it the second largest receipient country in the world after India (Banco de México, 2026b; World Bank, 2024). The U.S.–Mexico corridor is especially important because 97% of the migrant population in Mexico resides in the United States (Li Ng, 2024). This paper asks whether a negative local shock in the United States is transmitted to Mexican municipalities through the lens of remittances. Do migrants smooth transfers after being hit by a shock, or do origin households bear part of the adjustment through lower remittance inflows?

The answer matters because remittances are a significant part of household resources, with migrant households of Mexican origin residing in the U.S. sending on average around US$12,000 per year (Isidro Luna & Martínez Hernández, 2026). While our data does not directly observe Mexican household income, prior work shows that remittances in Mexico affect consumption, poverty, inequality, food security, schooling, and child labor decisions (Alcaraz et al., 2012; Mora-Rivera & van Gameren, 2021; Taylor et al., 2005). We therefore interpret changes in municipal remittance inflows as reduced-form evidence on a channel with direct relevance for household income and consumption, while not claiming to estimate those household outcomes themselves.

Our paper contributes to two distinct strands of the migration and remittance literature. First, we contribute to the literature on international shock transmission, for which the primary empirical difficulty lies in the fact that major economic crises typically affect both sides of a migration corridor. For instance, while Caballero et al. (2023) provide important evidence that U.S. labor market shocks during the Great Recession reduced remittances to Mexico, the global nature of that crisis affected both countries simultaneously. Furthermore, their study focuses on the extensive margin of remittance receipt. We instead choose to explore a more localized shock and provide estimates on the intensive margin. Specifically, we focus on Hurricane Ian, the costliest hurricane in Florida's history (Bucci et al., 2026), which struck solely the sender side of the corridor. Because natural disasters severely restrict local labor demand and household resources (Boustan et al., 2020), Hurricane Ian provides a clean, exogenous contraction to migrants' remitting capacity.

Second, we contribute to the literature studying remittances as a form of informal household insurance (Amuedo-Dorantes & Pozo, 2006). The vast majority of existing evidence focuses on shocks in the receiving country, demonstrating how migrants increase transfers to insure their families against local disasters at origin. We invert this framework by examining a sender-side shock, where the economic disruption occurs in the destination labor market while origin municipalities remain untouched.

Chapter 2 introduces a structural remittance model to formalize the impact of the shock on remittance flows. The model follows the insight provided by Albert and Monras (2022), who show that migrants allocate resources across where they live and where their origin households are located. In our setting, this idea is applied to remittance choices rather than to location choices, with migrants choosing between local consumption in the United States and resources available to the origin household in Mexico. They differ in income and in the strength of their altruistic or obligation motive, and remitting involves both a fixed cost and a variable cost. In the spirit of models with heterogeneous firms, fixed costs, and variable costs (Melitz, 2003), the fixed cost generates an extensive margin of remitting, while the variable cost governs the intensity of transfers. We then augment the model to account for the fact that origin-household resources include local non-remittance income plus remittances received from migrants in other U.S. states. This extension allows us to interpret both the direct exposure effect and possible spillover effects (Hudgens & Halloran, 2008).

Chapter 3 presents hurricane Ian as a source of exogenous variation in remittance flows, documenting its impact on Florida remittance outflows and labor sectors with high employment of Mexican migrants. Because bilateral flows between U.S. states and Mexican municipalities are not publicly available, we study aggregated municipality remittance inflows and construct a continuous treatment variable using Matricula Consular de Alta Seguridad (MCAS) administrative records. MCAS records identify the Mexican municipality of origin and U.S. state of residence of Mexican consular-card holders, enabling us to map migrant networks. We use these records to construct state-to-municipality migration shares, which we use as a proxy for exposure to each U.S. state, representing our treatment variable.

Chapter 4 studies the direct impact of the shock on remittance inflows to Mexican municipalities. To do so, we employ a continuous Difference-in-Differences (DiD) event-study design, where the treatment is the pre-shock exposure to Florida. We find that in the quarter of the shock, a one percentage point increase in Florida exposure is associated with a 0.16 percentate point decrease in remittance inflows. However, this point estimate is comparable in magnitude to some of the pre-trend fluctuations. Since there appears to be a cyclical component that is not fully captured by the fixed effects, this drop could partly reflect a cyclical fluctuation rather than the pure effect of the shock.

Building on the results of Chapter 4, Chapter 5 explores potential spillover effects from other U.S. states, which may be masking the true impact of the shock on remittance inflows to Mexican municipalities. We find statistically significant spillover effects from Texas, the second largest sender state, but not from California, the largest sender state. The spillover effect from Texas is negative, meaning that the more exposed a given municipality is to Texas, the more pronounced is the negative post-shock effect of the Florida exposure on remittance patterns. This finding suggests that, instead of a compensatory effect, we observe an amplification of the negative impact of the Florida shock on remittances in municipalities that are also highly exposed to Texas. One possible interpretation of this result is internal mobility within the U.S.. For municipalities with strong pre-existing connections to Texas, these established networks may allow migrants to easily relocate from Florida to Texas after the hurricane, cutting them out from the reconstruction boom that follows, and leading to a decrease in their income and therefore in their remittance capability.

2. Theoretical framework and empirical strategy

This chapter outlines the theoretical framework used for our analysis. We begin by adapting a heterogeneous-firm trade model to the context of individual remittance behavior, establishing the theoretical foundations of how a migrant's transfer decisions can be influenced by economic shocks. Then, we expand the model by allowing for spillover effects across migrant networks. Finally, we provide the main predictions of the model, which guide our empirical strategy in the subsequent chapters.

2.1. A Remittance Model with Heterogeneous Migrants

To analyze the impact of shocks on remittance flows, we adapt the logic of heterogeneous-firm trade models to the context of remittances. Specifically, we consider a model where migrants differ in income and altruism intensity, and where remitting requires paying both a fixed access cost and a variable cost proportional to the amount remitted. Formally, consider migrant i, who resides in U.S. state s, originates from Mexican municipality m, and chooses remittances in period t. The migrant's utility function is given by:

$$U_{imst} = \left[\alpha\left(c^L_{imst}\right)^{\frac{\sigma-1}{\sigma}} + (1-\alpha)\left(\phi_{im}\left[R_{imst} + y^O_{mt}\right]\right)^{\frac{\sigma-1}{\sigma}}\right]^{\frac{\sigma}{\sigma-1}} \tag{2.1}$$

Here $\phi_{im} > 0$ denotes the altruism or obligation intensity parameter, analogous to the productivity parameter in Melitz (2003). This strictly positive parameter ensures that sending remittances can increase the migrant's utility by raising origin-household resources. The variable $c^L_{imst} \geq 0$ represents the migrant's consumption in the destination country, and $R_{imst} \geq 0$ is the amount sent in remittances by migrant i. The weight of local consumption is given by $\alpha \in (0,1)$, and the elasticity of substitution between local consumption and origin-household resources is given by σ. Origin-household total resources are therefore defined as $R_{imst} + y^O_{imst}$, where the presence of y^O_{imst} introduces a role for origin-country economic conditions: lower perceived origin resources increase the marginal utility of remittances, generating an insurance motive consistent with Yang and Choi (2007).

The migrant's budget constraint is given by:

$$c^L_{imst} + f_{mt}\mathbb{1}\{R_{imst} > 0\} + (1+\tau_{mt})R_{imst} \leq w_{imst} \tag{2.2}$$

where $w_{imst} \geq 0$ is the migrant's wage, $f_{mt} \geq 0$ is a fixed remitting cost capturing the pre-transaction expense of accessing a remittance channel, analogous to the fixed export cost in Melitz (2003), and $\tau_{mt} > -1$ is the variable cost which scales with the amount remitted, analogous to the iceberg cost structure in Melitz (2003). Notice that, even though τ_{mt} is typically positive to reflect transaction fees or exchange rate spreads, we allow for the possibility of negative variable costs, which could arise from the presence of remittance subsidies or favorable exchange rate regimes.

If the migrant maximizes utility subject to the budget constraint and chooses to remit a positive amount ($R_{imst} > 0$), solving the Lagrangian yields the optimal remittance function:

$$R^*_{imst} = \frac{(w_{imst} - f_{mt}) - \frac{\phi_{im}}{A_{imst}} y^O_{mt}}{\frac{\phi_{im}}{A_{imst}} + (1 + \tau_{mt})} \tag{2.3}$$

where A_{imst} represents the optimal ratio of origin-household resources to own local consumption, and is defined as:

$$A_{imst} \equiv \left(\frac{(1 - \alpha)\phi_{im}}{\alpha(1 + \tau_{mt})} \right)^{\sigma} \tag{2.4}$$

To understand the effect of destination-country income shocks on remittance flows, we take the partial derivative of optimal remittances with respect to wage:

$$\frac{\partial R^*_{imst}}{\partial w_{imst}} = \frac{A_{imst}}{1 + (1 + \tau_{mt})A_{imst}} > 0 \tag{2.5}$$

Since the derivative is strictly positive, through an income effect the model predicts that a positive shock to the migrant's wage will increase remittances, while a negative shock will decrease them. The fixed cost f_{mt} establishes an entry threshold: if the migrant's wage is not sufficiently high relative to the fixed cost, they will not send remittances at all. Additionally, the variable cost τ_{mt} reduces the marginal benefit of each dollar sent, dampening the sensitivity of remittances to wage fluctuations.

2.2. Allowing for Spillovers

We expand the model to allow for spillover effects across migrant networks. Decomposing origin-household resources into local non-remittance income and remittances received from migrants in other US states, the perceived origin-household resources excluding migrant i's own remittances can be expressed as:

$$y_{imst}^{O} = \tilde{y}_{mt}^{O} + \sum_{s' \neq s} R_{s'mt} \tag{2.6}$$

Solving for optimal remittances in the augmented model yields:

$$R_{imst}^{*} = \frac{(w_{imst} - f_{mt}) - \frac{\phi_{im}}{A_{imst}} \left(\tilde{y}_{mt}^{O} + \sum_{s' \neq s} R_{s'mt}\right)}{\frac{\phi_{im}}{A_{imst}} + (1 + \tau_{mt})} \tag{2.7}$$

Hence, holding the migrant's own wage and remitting costs fixed, remittances from other migrants enter the decision through perceived origin-household resources:

$$\frac{\partial R_{imst}^{*}}{\partial \left(\sum_{s' \neq s} R_{s'mt}\right)} = -\frac{\phi_{im}}{\phi_{im} + A_{imst}(1 + \tau_{mt})} \tag{2.8}$$

Because the derivative is strictly negative, high exposure to a destination shock lowers perceived origin resources and raises the marginal utility of alternative transfers. This drives compensatory network spillovers, where the direct remittance drop generates an offsetting increase from other destinations.

To ground these concepts, the model delivers two empirical implications that guide our analysis. First, because optimal remittances are increasing in migrant income, a negative wage or income shock in the destination country should reduce remittances sent by affected migrants. Second, because remittances from other destinations enter the migrant's decision through perceived origin-household resources, a decline in transfers from the affected destination raises the marginal value of remittances from migrants in alternative destinations connected to the same municipality. Hence, origin municipalities with greater pre-existing exposure to the affected migrant destination should experience larger declines in total remittance receipts, all else equal. Based on these predictions, we develop our empirical strategy as follows: first, we test whether local inflows fall more in Mexican municipalities highly exposed to the destination hit by the shock; then, we examine whether co-exposure to other high-sending US states not directly affected mitigate the negative impact on total remittance revenue.

3. Empirical Context and Data

This chapter introduces the empirical context and data used in our analysis. We begin by introducing our source of exogenous variation in remittance flows, Hurricane Ian. First, we document the hurricane's impact on Florida's remittance outflows as well as sectors with high Mexican immigrant employment. Next, we explain the data sources and the methodology we use to construct our treatment measure, which differentiates Mexican municipalities by how exposed they are to the shock. Finally, we present a validation exercise, justifying the choice of our constructed treatment variable as a proxy for the Mexican migrant population in the United States.

3.1. The Natural Experiment: Hurricane Ian (2022)

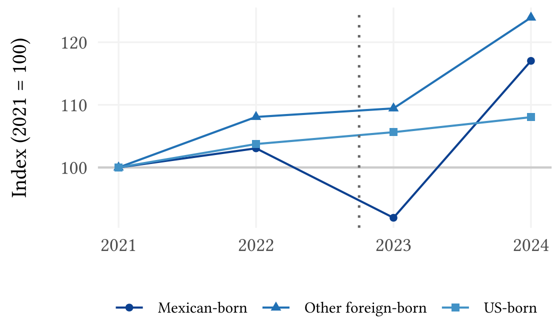
Hurricane Ian struck the southwestern coast of Florida on September 28, 2022, with peak sustained winds of 140 kt (around 260 kph). Ian caused an estimated US$ 112.9 billion in total damage in the United States, of which US$ 109.5 billion occurred in Florida alone, making it simultaneously the costliest hurricane in Florida's history and the third-costliest in US history. In Lee County alone, at least 52,514 structures were impacted, of which 5,369 were completely destroyed (Bucci et al., 2026).

Beyond its aggregate toll, Ian's destruction fell on sectors that employ a large share of Mexican migrants. Construction is the single largest sector of employment for Mexican-born workers in Florida, at roughly 28% of all those employed, and the four sectors most exposed to the storm together account for over 50% of Mexican workers. Within them, construction makes up 52%, agriculture 17%, landscaping 16%, and hospitality 15%.[1] Construction is therefore both the main source of work for Mexican workers and the channel where Ian's damage translates most directly into lost employment, and later into reconstruction demand.

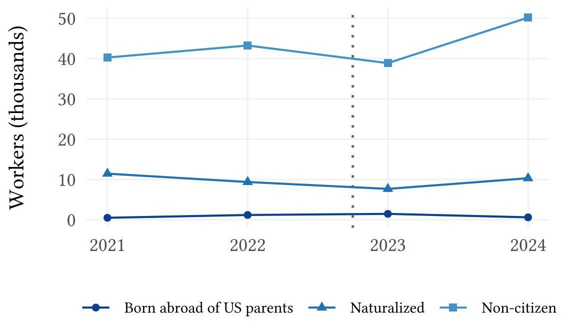
Figure 3.1a indexes Florida construction employment to 2021 by workers' region of birth. The Mexican-born series diverges from the US-born and other foreign-born series after

[1]Calculated using IPUMS USA, ACS 1-year samples 2021–2024, employed Mexican-born residents of Florida.

Ian, falling to roughly 92% of its 2021 level in 2023 before rebounding to about 117% in 2024. The US-born series drifts upward throughout, which suggests the disruption was specific to migrant labor rather than to construction as a whole. Figure 3.1b shows that this movement is carried by non-citizens, who make up the large majority of Mexican-born construction workers in the state and absorb both the 2023 dip and the 2024 rebound. Non-citizens are also the group most reliant on remittances as a financial link to their origin households (Handlin et al., 2002). The descriptive evidence thus indicates that Ian affected workers whose earnings finance remittances to Mexico, which motivates the migrant income channel we examine in the remainder of the paper.



(a) Construction employment by region of birth, indexed to 2021



(b) Mexican-born construction workers by citizenship status

Figure 3.1: Mexican-born construction employment in Florida around Hurricane Ian. Source: IPUMS USA, ACS 1-year samples 2021–2024; employed Florida construction workers. Dotted line marks Hurricane Ian (2022Q4).

Crucially for our identification strategy, Hurricane Ian did not affect Mexico. The storm originated in the Caribbean, made landfall in Cuba and Florida, and dissipated over the Atlantic after a final, less impactful landfall in South Carolina (Bucci et al., 2026). Therefore, the shock is spatially clean, allowing us to better isolate the transmission mechanism through the remittance channel. In the language of the model proposed in Chapter 2, Hurricane Ian is the negative shock to migrants' economic resources, Florida is the affected destination, and Mexican municipalities differ in their pre-existing exposure to Florida through migration networks.

To motivate the analysis, Figure 3.2 presents the evolution of remittance outflows from Florida to Mexico using data from Banxico series CE168 (Banco de México, 2026a). A sharp decline in remittance outflows can be observed in the quarter immediately following Hurricane Ian, with flows falling by approximately 40% relative to the previous quarter. This pronounced contraction suggests that the hurricane generated a substantial disruption in remittance activity from Florida to Mexico, thereby providing a strong motivation for examining its effects on Mexican municipalities' remittance inflows.[2]

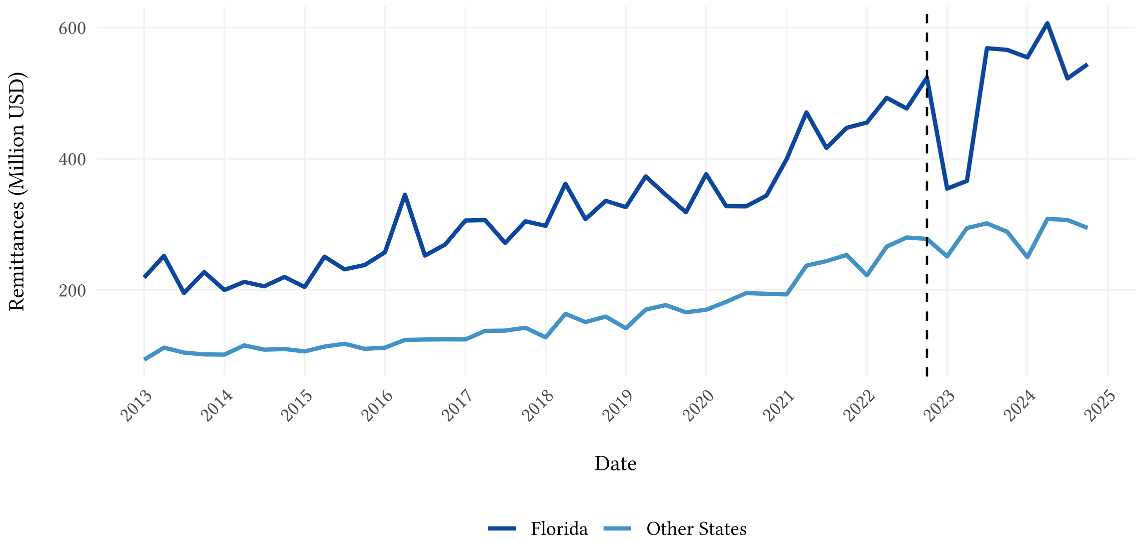


Figure 3.2: Remittance outflows from Florida to Mexico, in millions of US dollars. The vertical dashed line indicates the quarter of Hurricane Ian (Q4 2022).

3.2. Data Sources

Given that bilateral remittance flows between US States and Mexican municipalities are not publicly available, we base our estimation strategy on multiple data sources. The source of remittance data is Banco de México (Banxico), through the datasets provided by its Sistema de Información Económica (SIE). Specifically, series CE166 records remittance inflows received by each Mexican municipality. This series is available at quarterly frequency, from Q1 2013 to Q1 2026 and represents our outcome variable (Banco de México, 2026c). To capture the Mexican migration network, we utilize administrative records from the Matricula Consular de Alta Seguridad (MCAS) program, managed by the Instituto de los Mexicanos en el Exterior (IME) (Instituto de los Mexicanos en el Exterior, 2026). The publicly available version of this dataset provides a comprehensive registry of consular

[2]Florida outflows are plotted against the average outflows of all other U.S. states. Because the comparison series is constructed as an average, potential opposite shocks affecting individual states during the Hurricane Ian period may be attenuated. Nevertheless, the purpose of the figure is simply to illustrate the magnitude of the decline observed in Florida.

identification cards issued to Mexican nationals residing in the United States each year, from 2010 to 2024. While the MCAS does not constitute an exhaustive census of all migrants, it represents a robust proxy for the spatial distribution of Mexican migrants in the United States. In particular, it is highly representative of undocumented and recently arrived migrants, who face the strongest incentives to obtain consular identification (Caballero et al., 2018).

3.3. Migration Networks Construction

To build our treatment measure, we construct a migration-based matrix using annual MCAS data from 2014 to 2021. Because these records capture new consular issuances rather than the complete stock of the remitting population, which also includes previously arrived migrants, we rely on the assumption of persistence of migration patterns over time (Card, 2001; Munshi, 2003). Additionally, we rely on the fact that there is no significant internal mobility within Mexico, allowing us to link remittance receiving municipalities to the migrants' origin villages. This is supported by the literature, showing that internal mobility has decreased over the last few decades (Gordillo de Anda & Plassot, 2017; Partida Bush, 2014).

To mitigate the volatility of yearly migration fluctuation and construct a robust proxy for the long-term spatial distribution of Mexican migrants, we compute the average of the migration matrices across all years. Formally, we define $MCAS_{ms}$ as the average annual count of MCAS issuances to migrants from Mexican municipality m residing in U.S. state s. For each Mexican municipality, we calculate the average migration network share, w_{ms}, which represents the proportion of municipality m's total U.S. migrant population that resides in state s. This is calculated as the ratio of $MCAS_{ms}$ to the total number of MCAS issuances for migrants from municipality m across all U.S. states S:

$$w_{ms} = \frac{MCAS_{ms}}{\sum_{s=1}^{S} MCAS_{ms}} \tag{3.1}$$

By construction, the sum of w_{ms} across all U.S. states for a given Mexican municipality m equals 1, and the value of w_{ms} for a specific state s reflects the relative importance of that state as a destination for migrants from municipality m, which in turn is a proxy for the exposure of municipality m to shocks occurring in state s.

3.4. Persistence Assumption and Validation

The validity of this estimation relies on the assumption of temporal persistence of migration flows, implying that the distribution of migrants recorded in the MCAS data is representative of the active remitting population. The stability of migration patterns over time is well-documented in the literature, with studies showing that existing migration networks tend to dictate future migration patterns, creating persistent flows between specific origin and destination pairs (Card, 2001; Munshi, 2003). To verify this assumption within our specific sample period, we conduct a validation exercise using MCAS data. Figure 3.3 computes the Pearson correlation of our constructed weighting matrices across years. As observed, when taking any year from 2014 onward as the base year, the correlation with subsequent years is consistently above 0.90.

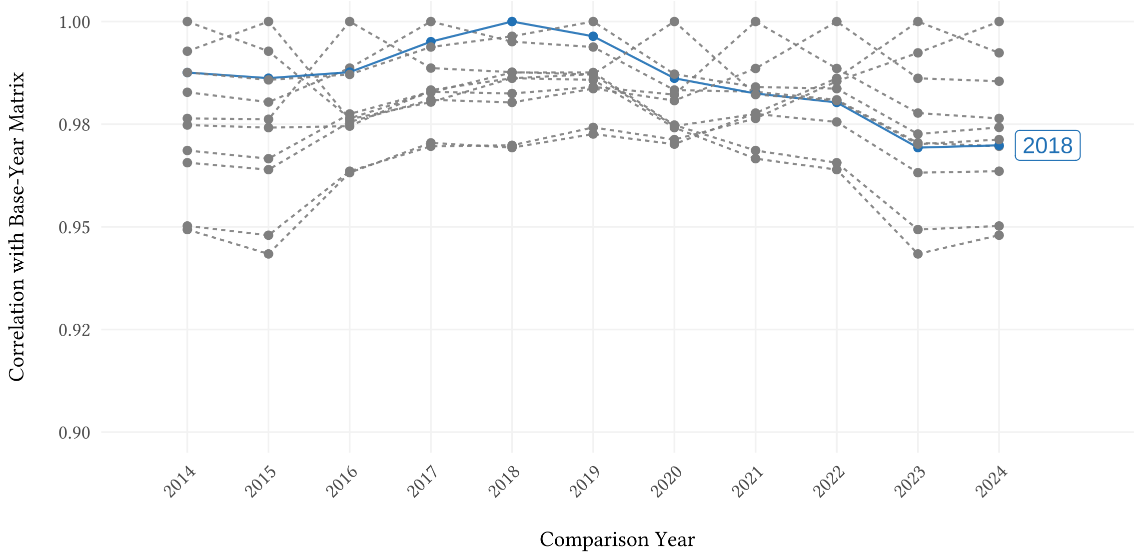


Figure 3.3: Correlation of MCAS-based weighting matrices across years. Each point represents the correlation coefficient between the weighting matrix of the base year and the weighting matrix of the comparison year (2018 as base year is highlighted in blue for demonstration purposes).

This high degree of correlation suggests that the spatial distribution of Mexican migrants across U.S. states has remained highly stable over time, lending credibility to our use of an averaged weighting matrix for estimating bilateral remittance flows. Given that Hurricane Ian hit Florida in Q4 2022 and the MCAS data for Florida in 2013 is missing, to construct our treatment variable we use the average of the weighting matrices from 2014 to 2021, which are all highly correlated with each other, and avoid any endogeneity in the treatment caused by the shock.

4. Empirical Methodology and Results

This chapter presents the core empirical strategy and the main estimation results for the Florida shock. We explain why a continuous DiD event-study design is appropriate when treatment intensity varies across municipalities, and argue for our choice of PPML as a valid and robust approach for handling our remittance data. Then, we show the dynamic effect of Florida exposure on municipal remittance inflows and test for robustness by trying alternative clustering specifications.

4.1. Estimation Strategy

The primary challenge in estimating the impact of the Hurricane Ian shock on remittance inflows to Mexican municipalities lies in the sorting of migrants. Municipalities with higher exposure to Florida may differ systematically from those with networks concentrated in other U.S. states. Therefore, a standard cross-sectional analysis comparing municipalities with different levels of exposure to Florida would likely lead to biased estimates due to confounding factors correlated with both the treatment (exposure to Florida) and the outcome (remittance inflows). This classic omitted variable bias problem makes a causal interpretation of cross-sectional estimates impossible (Wooldridge, 2010). To overcome this challenge, we leverage the temporal variation in remittance inflows by employing a continuous Difference-in-Differences (DiD) event-study design (Callaway et al., 2024), where our treatment is the pre-shock exposure to Florida, measured by $w_{m,\text{FL}}$.

4.2. Event-Study Specification

We use the following Poisson Pseudo Maximum Likelihood (PPML) event-study specification to analyze the impact of the Hurricane Ian shock on remittance inflows to Mexican municipalities:

$$\mathbb{E}[R_{mt} \mid \mathbf{X}_{mt}] = \exp\left(\alpha_m + \gamma_{s(m),t} + \sum_{k \neq -1} \beta_k \cdot \left(w_{m,\text{FL}} \cdot \mathbb{1}\{t - T_0 = k\}\right)\right) \tag{4.1}$$

The event-study specification allows us to estimate the dynamic treatment effects of the shock over time. We include municipality fixed effects (α_m) to control for all time-invariant, unobserved municipal characteristics that might otherwise introduce omitted variable bias. Additionally, we incorporate state-by-time fixed effects ($\gamma_{s(m),t}$), where $s(m)$ denotes the specific Mexican state to which municipality m belongs. These fixed effects account for both macroeconomic shocks common to all regions and time-varying unobservables at the state level. The interaction term $w_{m,\text{FL}} \cdot \mathbb{1}\{t - T_0 = k\}$ captures the differential effect of the shock on Mexican municipalities based on their varying degrees of baseline exposure to Florida, where $w_{m,\text{FL}}$ is a measure of pre-shock exposure to Florida and $\mathbb{1}\{t - T_0 = k\}$ is an indicator for each time period relative to the shock. Therefore, the β_k coefficients represent the change in remittance inflows for a one-percent increase in Florida exposure at each time period k relative to the shock. Errors are clustered at the municipality level to account for potential correlation within municipalities over time.

4.3. Robustness Checks

Following Silva and Tenreyro (2006), we employ the Poisson Pseudo Maximum Likelihood (PPML) estimator to address the heteroskedasticity and excess zeros, which are common features of remittance data at the municipality level. Unlike log-linear models, PPML yields consistent estimates regardless of heteroskedasticity, relying only on the correct specification of the conditional mean. However, heteroskedasticity affects inference, which we handle by clustering standard errors by municipality. This narrows our robustness checks to just two fundamental issues: the specification of the exponential conditional mean, and the data's mean-variance structure.

We address the first with a Ramsey RESET test as proposed by Ramsey (1969) and applied to this setting by Silva and Tenreyro (2006), which augments each specification with the square and cube of its fitted values and tests their joint nullity. A rejection signals that the assumed conditional mean is misspecified. We find that a log-linear OLS specification strongly rejects correct specification at all conventional levels ($W = 835.10, p < 0.001$), whereas the PPML specification does not reject the null ($W = 1.18, p = 0.308$).

The second question is answered by Figure 4.1, which plots, for each municipality, the variance of its pre-shock quarterly remittance inflows against their mean, both in logs.

On these axes the Poisson benchmark is the line of slope one, along which the variance rises one-to-one with the mean. The cloud lies well above this line and follows a slope of approximately 1.546, so the variance scales faster than the mean, signaling overdispersion. This leaves PPML unaffected, since its consistency does not depend on the variance function. A log-linear model performs worse: the expected value of log inflows is not a linear function of the regressors, so estimating the model in logs by OLS does not recover the parameters of the conditional mean. The two diagnostics thereby support PPML as the baseline estimator.

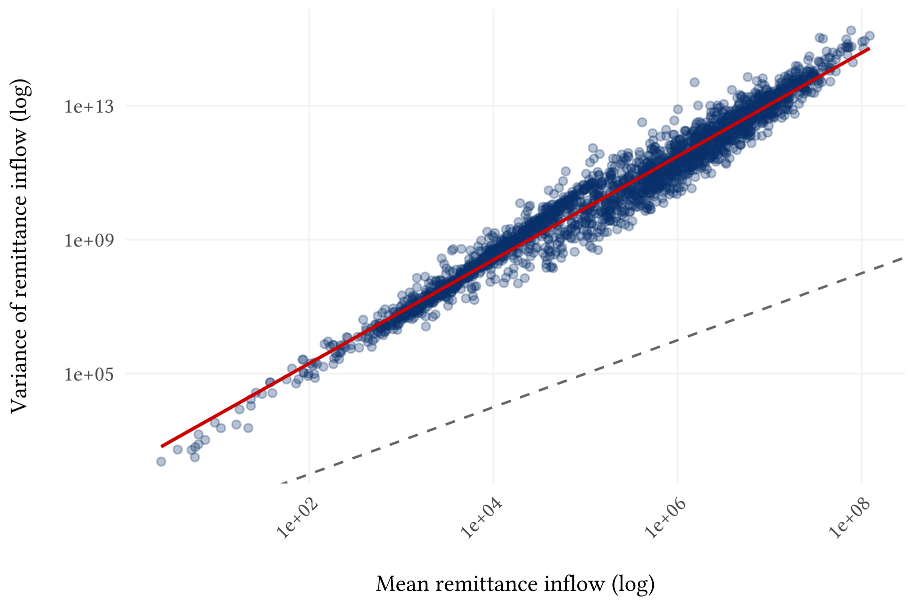


Figure 4.1: Mean-variance relationship of municipality remittance inflows. Each point represents a Mexican municipality; axes show the variance against the mean of its quarterly inflows over the pre-shock window. The solid line is the fitted relationship; the dashed line marks the Poisson equidispersion benchmark, where variance equals the mean.

4.4. Results

Figure 4.2 shows the result of the PPML event-study estimation, where the vertical dashed line represents the timing of the shock (Q4 2022). The pre-shock coefficients (β_k for $k < 0$) are close to zero and statistically insignificant, visually consistent with the parallel trends assumption underlying our DiD design. The point estimate in the quarter of the shock (Q4 2022) is negative and statistically significant at the 1% level, indicating that municipalities with higher exposure to Florida experienced a larger drop in remittance inflows immediately following the shock. Specifically, a one percentage point increase in Florida exposure is associated with a 0.16 percentage point decrease in remittance inflows in the quarter of the shock. In the subsequent quarters, the coefficients become slightly positive but statistically insignificant at the 5% level, suggesting that the initial negative impact of the shock on remittance inflows dissipated over time.[1]

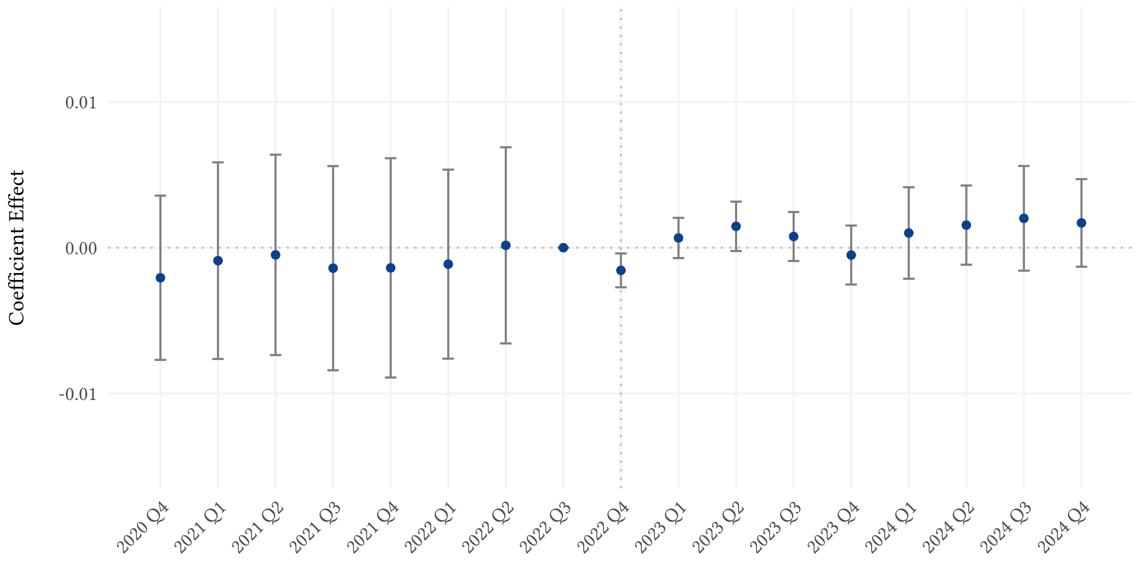


Figure 4.2: PPML event study estimates of the impact of Florida exposure on remittance inflows. Each point represents the estimated coefficient β_k for the interaction between Florida exposure and the time period k relative to the shock, with 95% confidence intervals. The vertical dashed line indicates the timing of the shock (Q4 2022).

However, the absolute magnitude of the point estimate in Q4 2022 is comparable to some of the pre-trend fluctuations. Notably, there is a cyclical component that is not fully captured by the fixed effects, meaning that this drop could partly reflect a cyclical fluctuation rather than the pure effect of the shock. Nevertheless, the confidence intervals shrink noticeably during the shock period, suggesting that the hurricane injected a clear signal into the data. Moreover, this drop in our PPML estimates precedes the contraction observed in aggregate Florida remittance outflows, which do not visibly decline until Q1 and Q2 of 2023. Given the complexity of migration networks—and the possibility that workers in other US states altered their behavior to compensate for the Florida shock—these initial municipal-level results remain somewhat ambiguous. To disentangle these dynam-

[1] We assess robustness to violations of parallel trends using the relative-magnitudes approach of Rambachan and Roth (2023), reporting the breakdown value $\bar{M}$ (the largest pre-to-post violation ratio for which the effect stays significant; $\bar{M} = 1$ is a conventional lower bound). The baseline Florida-only effect breaks down at $\bar{M} = 0.15$. The sign and timing of the response are thus credible, but its magnitude is fragile, as expected given the dispersed, complex network of remittance flows, which motivates the spillover analysis that follows.

ics, Chapter 5 explores the presence of spillover effects from other states, which may be masking the true impact of the shock on remittance inflows to Mexican municipalities.

As a robustness check, we also estimate the same event-study specification using spatially clustered errors, following the methodology of Conley (1999), to account for potential spatial correlation in remittance inflow shocks affecting neighboring municipalities. The results, shown in Figure 4.3, are qualitatively similar to the baseline specification, with the main difference being the tighter confidence intervals in the pre-period as buffer distance increases, and slightly wider in the post-shock period. Given that the results are robust to both clustering methods, and that the spatial clustering specification fails to produce positive definite variance-covariance matrices for buffer distances above 50km, we consider the municipality-level clustering as our baseline specification.

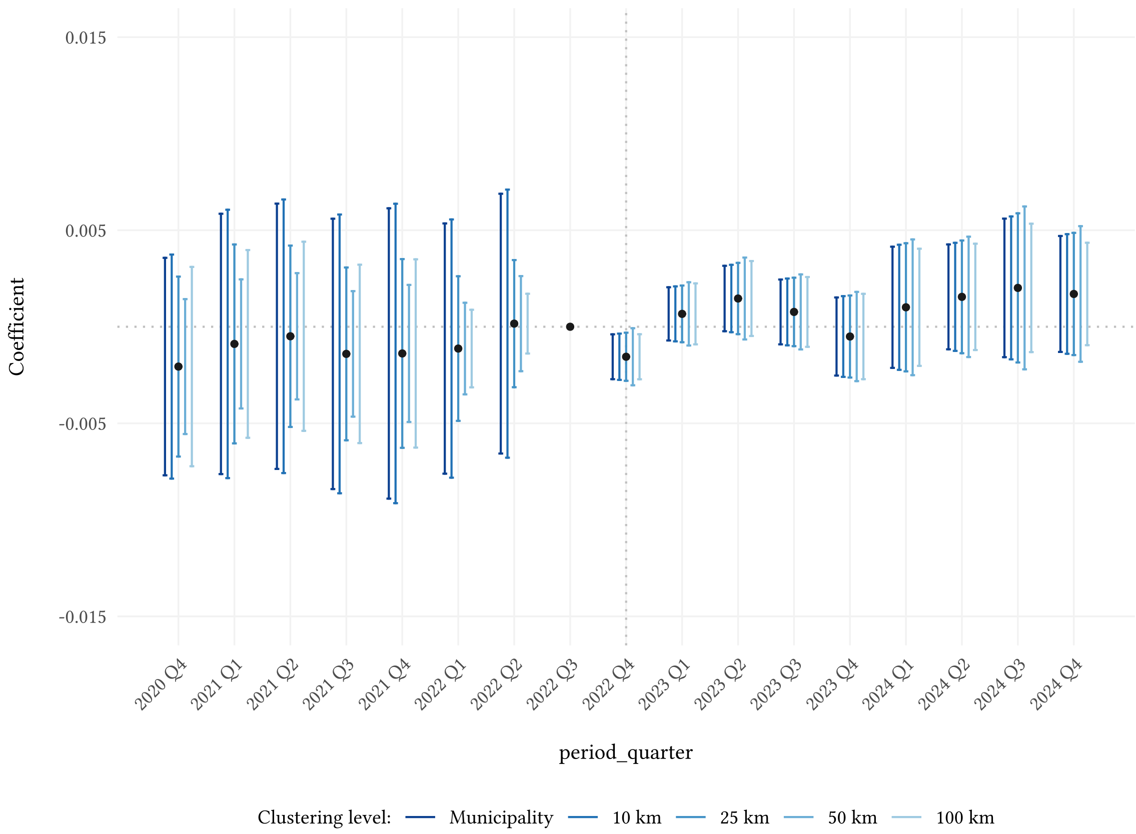


Figure 4.3: PPML event study estimates with Conley spatially clustered Standard Errors for different buffer distances.

5. Investigating Spillover Effects

This chapter examines whether the Florida shock affected Mexican municipalities not only through direct Florida exposure, but also through indirect links with Texas and California migrant networks. We build on the exposure-mapping logic from Aronow and Samii (2017) and evaluate whether these indirect networks offset or amplify the estimated effect of the Florida shock. We find economically plausible spillover effects from Texas that magnify the negative effect of the shock on remittance patterns.

5.1. Motivation

A fundamental requirement for standard causal inference models is the Stable Unit Treatment Value Assumption (SUTVA), formalized by Rubin (1980), which assumes that the potential outcomes of any unit depend exclusively on its own treatment assignment, excluding the possibility of any indirect spillover between units. In our context, U.S. states not directly exposed to a local shock might still be affected if control units are exposed to the treatment through indirect channels, leading to biased estimates (Aronow & Samii, 2017; Clarke, 2017). For instance, given that California and Texas all host a big proportion of Mexican migrants (each representing an average of 30% and 20% from the total, respectively), we hypothesize that these two states might have played a role in offsetting the negative Florida shock. If a Mexican municipality highly dependent on Florida remittances experiences an income shock due to Hurricane Ian, connected migrants residing in Texas might respond by increasing their transfers to compensate for the drop. If this was the case, the estimated Florida-exposure effect on total remittances would capture the net effect of the shock after within-network adjustment, rather than the direct fall in Florida-origin remittances.

5.2. Empirical Strategy

We test a possible compensation for the Florida shock from the states of Texas (TX) and California (CA) expanding the previous PPML specification with an interaction term:

$$\mathbb{E}[R_{mt} \mid \mathbf{X}_{mt}] = \exp\left(\alpha_m + \gamma_{s(m),t} + \sum_{k \neq -1} \mathbb{1}\{t - T_0 = k\} \cdot \left[\beta_{1,k} w_{m,\mathrm{FL}} + \beta_{2,k} w_{m,\mathrm{TX}} + \beta_{3,k} (w_{m,\mathrm{FL}} \times w_{m,\mathrm{TX}})\right]\right)$$

Consistent with the specification of Chapter 4, the equation above accounts for municipality fixed effects (α_m), Mexican state-quarter fixed effects ($\gamma_{s(m),t}$) and clustered standard errors at the municipality level. Note that an analogous specification can be estimated for the CA state, by replacing $w_{m,\text{TX}}$ with $w_{m,\text{CA}}$.

To understand the mechanics of this spillover model, we can isolate the post-shock marginal effect of Florida exposure on expected municipal remittance inflows. Taking the partial derivative of the log conditional expectation with respect to Florida exposure yields:

$$\frac{\partial \ln \mathbb{E}[R_{mt} \mid \mathbf{X}_{mt}]}{\partial w_{m,\text{FL}}}\bigg|_{\text{Post} = 1} = \beta_1 + \beta_3 w_{m,\text{TX}} \tag{5.1}$$

Mathematically, β_1 represents the percentage change in remittances resulting from a one-unit increase in Florida exposure, evaluated strictly where Texas exposure is zero ($w_{m,\text{TX}} = 0$). On the other hand, the coefficient of the interaction term, β_3 measures how the post-shock effect of Florida exposure changes for every one percentage point increase in Texas exposure. Intuitively, if $\beta_3 > 0$, the Texas network compensates for the Florida's drop in remittances, mitigating the negative impact of the shock. Conversely, if $\beta_3 < 0$, the Texas network amplifies the negative effect of the Florida shock, potentially due to out-migration costs.

Because β_1 measures how the hurricane impacted municipalities that rely on Florida but possess no alternative migration network in Texas, which might not represent a meaningful real-world baseline for many municipalities, we recenter the $w_{m,\text{TX}}$ variable at different percentiles of exposure to Texas. For example, if we recenter Texas exposure at its 90th percentile, the newly estimated β_1 gives us the isolated impact of the Florida shock on highly networked municipalities. By comparing the β_1 estimates across models recentered at low versus high percentiles of Texas exposure, we can formally test how the presence of a secondary network alters the impact of the Florida shock on remittance inflows.

5.3. Results

Table 5.1 reports the post-shock estimates of the β_3 interaction coefficient quarter by quarter for the Texas (TX) and California (CA) specifications.

Table 5.1: β_3 Coefficient Event-Study Estimates (Pre and Post Periods). *Note: Coefficients and standard errors are multiplied by 100 to represent percentage effects. Significance levels: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.*

Quarter	FL × TX		FL × CA	
2020 Q4	0.01	(0.01)	-0.00	(0.02)
2021 Q1	-0.00	(0.02)	0.02	(0.02)
2021 Q2	0.00	(0.02)	0.01	(0.01)
2021 Q3	0.01	(0.01)	0.02	(0.01)
2021 Q4	0.01	(0.02)	0.02	(0.02)
2022 Q1	0.00	(0.01)	0.02	(0.01)
2022 Q2	-0.00	(0.01)	0.01	(0.01)
2022 Q4	0.01	(0.01)	0.00	(0.00)
2023 Q1	-0.02***	(0.01)	-0.00	(0.01)
2023 Q2	-0.02***	(0.01)	-0.01	(0.01)
2023 Q3	-0.02**	(0.01)	0.00	(0.01)
2023 Q4	-0.01	(0.01)	-0.00	(0.01)
2024 Q1	-0.04**	(0.01)	-0.01	(0.01)
2024 Q2	-0.04**	(0.01)	-0.00	(0.01)
2024 Q3	-0.05***	(0.01)	-0.00	(0.01)
2024 Q4	-0.05***	(0.01)	-0.01	(0.01)
Observations	41,327			
Pseudo R^2	0.989			
Municipality FE	Yes			
State × Quarter FE	Yes			
SE Cluster	Municipality			

In the CA case, the coefficients remain close to zero throughout the post-shock period, which suggests that no meaningful spillover effects are detected. By contrast, the TX coefficients become negative, statistically significant, and increasingly persistent over time, meaning that the more exposed a given municipality is to TX, the more pronounced is the negative post-shock effect of FL exposure on remittance patterns. In other words, higher TX exposure is associated with a larger net decline in remittances after the FL shock.

Since this result a priori does not support the compensatory spillover hypothesis of the model, we next aim to explore how the post-shock FL effect on remittances varies according to different levels of TX exposure. Figure 5.1 plots the β_1 coefficient, constructed by recentering the values of $w_{m,\mathrm{TX}}$ on the 10th and 90th percentiles of the distribution of TX exposure across municipalities. The negative net effect of the FL shock on remittances is stronger for municipalities with higher levels of TX exposure, which is consistent with the negative and statistically significant β_3 coefficient found above. This reinforces the idea that instead of a compensatory effect, we observe an amplification of the negative impact of the FL shock on remittances is more pronounced in municipalities that are also highly exposed to TX.

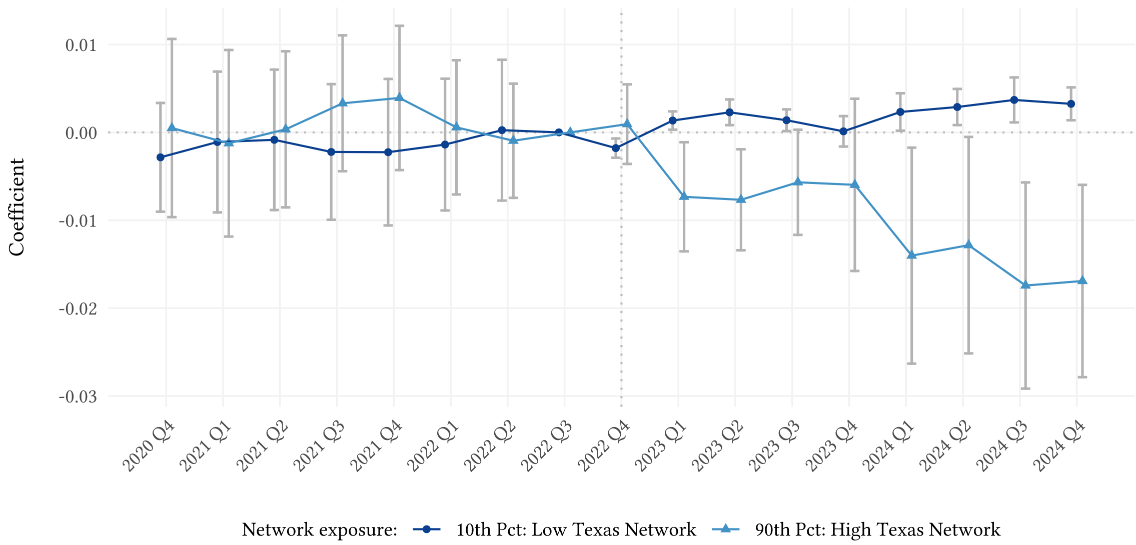


Figure 5.1: Net Florida shock evaluated at low and high levels of Texas exposure. The figure compares the implied post-shock Florida effect for municipalities at the 10th and 90th percentiles of Texas exposure.

A possible interpretation at the 90th percentile of Texas exposure is internal mobility within the US. For municipalities with strong pre-existing connections to Texas, these established networks may allow migrants to easily relocate from Florida to Texas after the hurricane. This behavior aligns with Fussell et al. (2023), who document immediate and persistent out-migration from disaster zones following Hurricanes Katrina and Rita. In contrast, migrants from municipalities at the 10th percentile of Texas exposure may find it much harder to relocate, thus remaining in Florida. Paradoxically, Figure 5.1 suggests that those who stay in Florida after the hurricane can participate in the reconstruction boom that follows, experiencing a wage increase and therefore an increase in remittances from the lense of our model discussed in Chapter 2. Conversely, those who relocate to Texas might face a decrease in their income due to frictions in the employment process or relocation costs.

This dynamic explains the timing observed in our event-study estimates. For municipalities highly connected to Texas, remittance inflows do not drop significantly in the exact quarter of the shock. Instead, the relative decline emerges and deepens in the subsequent quarters, particularly when compared to low-exposure municipalities. This delayed drop is consistent with the frictions of the migration process, while the delayed increase for the low-exposure municipalities reflects the participation in the reconstruction boom.

6. Conclusion

This study investigates the impact of a localized shock, Hurricane Ian, on remittance flows from U.S. states to Mexican municipalities and develops a conceptual framework to support the analysis. The baseline analysis reveals that a 1 percentage point increase in Florida exposure is associated with a 0.16 percentage points immediate decline in remittance inflows. However, potential compensatory effects may confound these results. While co-exposure with California yields no significant spillover effects, co-exposure with Texas generates a significant amplification effect. Indeed, a 1 percentage point increase in Texas exposure exacerbates the negative impact of the Florida shock. Specifically, municipalities highly exposed to Texas experience a larger decline in remittances in the quarters following the shock, whereas municipalities with low exposure show a small increase. This result contrasts with the mechanism implied by our model under compensatory behavior. One plausible explanation is that a strong Texas network enables easier out-migration from Florida. By relocating, these migrants forfeit the higher wages of Florida's post-hurricane reconstruction boom, leading to lower baseline incomes and thereby reducing their remittance capability.

Furthermore, our findings demonstrate that the effects on remittance flows from macro-level shocks, such as the Great Recession (Caballero et al., 2023), cannot directly be generalized to localized events. Localized shocks like Hurricane Ian disrupt only a specific subset of the population and have a limited impact on the broader U.S. economy. This allows for compensatory behavior, as unaffected migrants across the country can alter their remittance patterns to offset the drop in Florida remittances during the quarter of the shock or use their networks to absorb displaced workers. Our theoretical framework helps rationalize these findings through key parameters affecting remittance behavior, particularly the elasticity of substitution between migrants' own consumption and remittances, as well as the degree of altruism toward recipients. These network dynamics could be further explored by studying other localized economic or climate shocks or estimating the structural parameters. Finally, future research could substantially improve upon these results by utilizing bilateral state-to-municipality remittance data, which would allow researchers to explicitly separate the migration channels directly impacted by a shock from those that remain unaffected.

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